DHANASHALINI S

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with a B.Tech in Information Technology and experience building backend systems, applied-AI solutions, and full-stack applications in Python, TypeScript and Node.js. Delivered production-grade REST APIs, distributed job queues, retrieval-augmented generation (RAG) pipelines, and deep learning models — with measurable outcomes: 25–28% ML performance gains, 40% accuracy improvement, and 60% reduction in failure-triage time. Writes clean, tested, documented code; owns work end to end; thrives in Agile environments; and picks up new technologies quickly.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Backend & APIs: Node.js, Express, FastAPI, REST API design, BullMQ, async workers, concurrency, rate limiting, JWT authentication, Prisma ORM

AI / ML / Data: RAG, LLM APIs, prompt engineering, embeddings, vector search, reranking, PyTorch, TensorFlow, scikit-learn, CNN, U-Net, Pandas, Polars, NumPy, Explainable AI (SHAP, LIME)

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, responsive design

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis; database design, SQL query optimisation, data modeling

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, CI/CD, AWS EC2, AWS S3, Linux, Git, Postman

SRE & Automation: Site Reliability Engineering (SRE), monitoring & observability, alerting, SLIs/SLOs/SLAs, incident response, error budgets, fault-tolerance testing; test automation (pytest), automated regression gates, CI/CD pipeline automation, Python scripting, workflow automation, performance benchmarking

SOFT SKILLS

Problem Solving & Analytical Thinking • Communication & Documentation • Ownership & Accountability
Collaboration & Adaptability • Attention to Detail & Quality • Quick Learner • Agile Teamwork

PROFESSIONAL EXPERIENCE

Turing — Senior AI Trainer & Illustrator *Sep 2026 – Present*

- Evaluate AI-generated outputs daily against structured quality criteria, identifying logical and consistency defects and delivering detailed written feedback that feeds directly into model improvement cycles.
- Handle multi-domain annotation tasks (code, sketch, instruction-following) requiring precision, documentation, and systematic quality control across evolving project requirements.

Truerize IQ Strategic Solutions Pvt. Ltd. — Software Engineer Intern *Mar 2026 – May 2026*

- Raised land-cover classification accuracy 28% by building CNN and U-Net semantic-segmentation pipelines deployed as FastAPI REST services with SHAP and LIME explainability on satellite imagery.
- Cut batch and real-time inference latency 25% by replacing per-image Python loops with vectorised Polars and PyArrow columnar feature-extraction pipelines, documented with before/after benchmarks.

PROJECTS

CodeLens — AI Code Review & Repository Intelligence Platform *github.com/dhanashalini25/codelens*

Python · FastAPI · GitHub Actions · NumPy · SQLite · LLM APIs

- Cut PR review turnaround to under 90 seconds with a GitHub Actions CI pipeline running 13 AST-based static rules before the LLM pass, blocking committed credentials and shell-injection risks; 48 pytest tests green on Python 3.10–3.12.
- Improved code-search top-1 accuracy 40% via a NumPy hashed TF-IDF index with symbol-name weighting; benchmarked and documented trade-offs against BM25 (2 points worse).

Ibid — Grounded Document Question Answering (RAG) *github.com/dhanashalini25/ibid*

Python · FastAPI · Retrieval-Augmented Generation · Embeddings · SQLite

- Achieved 100% refusal accuracy on unanswerable queries via retrieval-score threshold + NO_ANSWER contract; lifted recall@5 from 95% → 100% (MRR 0.950 → 0.967) through keyword-overlap reranking, enforced by a 26-question CI harness.

PulseQueue — Distributed Background Job Processing Platform *github.com/dhanashalini25/pulse-queue*

Node.js · Express · BullMQ · Redis · Docker · REST APIs

- Sustained job delivery through worker crashes via exponential-backoff retries across 4 priority tiers and a dead-letter queue; reduced failure-triage time 60% via a real-time monitoring dashboard; API and worker tiers scaled independently via Docker Compose.

DevHire — Full-Stack Developer Hiring Platform *github.com/dhanashalini25/devhire-ai-hiring-platform*

Next.js · TypeScript · Node.js · Express · Prisma · PostgreSQL · JWT

- Built end-to-end full-stack app with JWT role-based auth (2 workflows), eliminated 100% of duplicate applications via PostgreSQL uniqueness constraints, and cut list-endpoint payloads 70% with server-side filtering and pagination.

EDUCATION

B.Tech, Information Technology — Knowledge Institute of Technology, 2025 | CGPA 7.8/10

B.A., Economics — Tamil Nadu Open University | Third University Rank

CERTIFICATIONS

Google Data Analytics Professional Certificate — Coursera

Google Cybersecurity Professional Certificate — Coursera